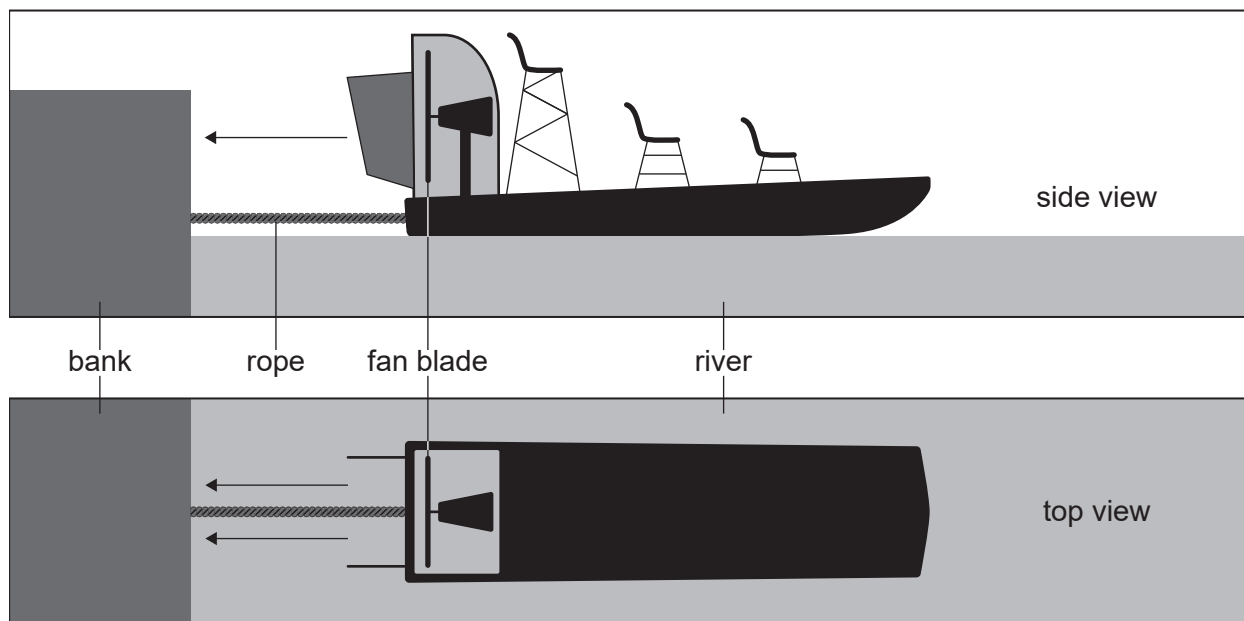


6. Airboats are used for transport across a river. To move the boat forward, air is propelled from the back of the boat by a fan blade.



An airboat has a fan blade of radius 1.8 m. This fan can propel air with a maximum speed relative to the boat of 20 m s^{-1} . The density of air is 1.2 kg m^{-3} .

- (a) Outline why a force acts on the airboat due to the fan blade.

[3]

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(This question continues on the following page)

(Question 6 continued)

- (b) In a test the airboat is tied to the river bank with a rope normal to the bank. The fan propels the air at its maximum speed. There is no wind.

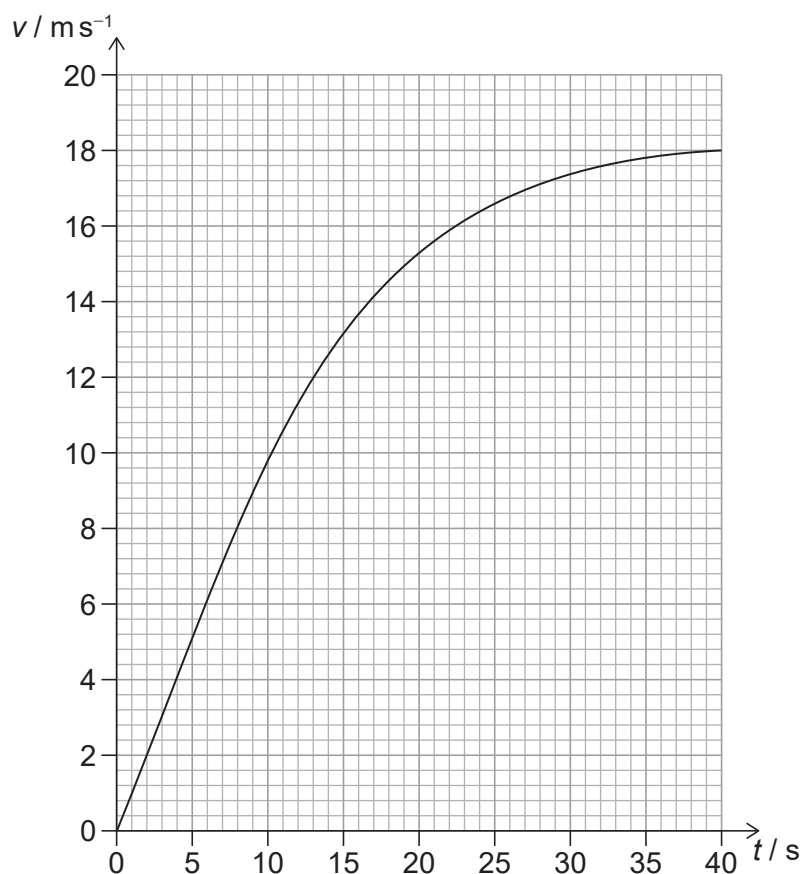
- (i) Show that a mass of about 240 kg of air moves through the fan every second. [2]

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- (ii) Show that the tension in the rope is about 5 kN. [1]

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- (c) The rope is untied and the airboat moves away from the bank. The variation with time t of the speed v of the airboat is shown for the motion.



(This question continues on the following page)

(Question 6 continued)

- (i) Explain why the airboat has a maximum speed under these conditions. [2]

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- (ii) Estimate the distance the airboat travels to reach its maximum speed. [2]

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- (iii) Deduce the mass of the airboat. [3]

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